

DATA VISUALIZATION AND ANALYTICS

Objective: To understand the importance of data visualization in analytics. To learn visualization techniques using Python libraries and spreadsheet tools. To create effective visualizations for data-driven decision-making.

Credit: 03

L–T–P: 3–0–0

Module No.	Content	Teaching Hours
I	Introduction to Data Visualization- Definition and importance of data visualization, Principles of effective visualization, Types of charts and graphs, Choosing the right visualization for data Data Visualization using Spreadsheets: Data Generation, Collection, Storage, Processing, and Analysis, The Role of a Data Engineer. Introduction to Excel / Google Sheets interface, Importing and cleaning data in spreadsheets, Basic charts: Column, Line, Pie, Bar, Advanced charts: Combo charts, Pivot charts Conditional formatting and sparklines Creating dashboards using PivotTables and Slicers Data Collection: Sources of Data, Data Lake, Batch Processing, Streaming, Web Scraping, APIs and Data Extraction. Data Modelling: Concepts, E-R Diagrams, Normalization and denormalization, OLAP, OLTP Data Storage: Relational Databases. NoSQL Databases. Data Warehouses. Data Processing and Transformation: ETL (Extract, Transform, Load) Processes	20
II	Introduction to Python for Data Visualization Python environment setup (Anaconda, Jupyter, Notebook, Google Colab), Basics of Python for data (NumPy, Pandas), Importing datasets (CSV, Excel, JSON), Data cleaning and preprocessing with Pandas Advanced Visualization Techniques: -Interactive visualizations using Plotly and Bokeh, Geospatial visualizations (optional), Visual storytelling with data. Exploratory Data Analysis (EDA): -Using visualizations to explore datasets, Identifying patterns, trends, and outliers, Case studies on real datasets. Data Cleaning Techniques- Batch Processing vs. Stream Processing Data Integration: Combining Data from Multiple Sources. Data Integration Tools. Data Quality and Consistency Visualizing Big Data: -Techniques and tools for handling and visualizing large datasets, ensuring performance and clarity. (Cloud-based visualization (e.g., AWS Quick Sight). Data Visualization with Matplotlib, Seaborn and Plotly - Matplotlib: Line plots, Bar charts, Histograms, Scatter plots, Customizing plots (labels, titles, Colores, legends, subplots), Seaborn: Statistical visualizations (count plot, boxplot, violin plot, heatmap), Pair plots and correlation visualization.	20

Textbooks:

- Data Engineering with Python by Paul Crickard.
- Fundamentals of Data Engineering by Joe Reis and Matt Housley.
- Data Visualization with Python and JavaScript by Kyran Dale

Reference Books:

- The Data Warehouse Toolkit: The Definitive Guide to Dimensional modelling by Ralph Kimball and Margy Ross.
- Designing Data-Intensive Applications by Martin Kleppmann.
- Big Data: Principles and Best Practices of Scalable Real-Time Data Systems by Nathan Marz and James Warren.
- VanderPlas, J. *Python Data Science Handbook*
- Knafllic, C.N. *Storytelling with Data*
- Excel/Google Sheets Documentation
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Outcome:

Upon completion of the course, students will:

- CO1: Understand the basics and importance of data engineering.
- CO2: Learn different data models and schema design techniques.
- CO3: Gain proficiency in data integration and ETL processes.
- CO4: Explore and use various big data technologies and tools.
- CO5: Understand the basics and importance of data visualization.

COs	POs/PSOs
CO1	PO1, PO2 / PSO1
CO2	PO3, PO4 / PSO2
CO3	PO5, PO6 / PSO3
CO4	PO1, PO3 / PSO1
CO5	PO2, PO4 / PSO2
CO6	PO7, PO8 / PSO4